Supplementary Material: Detailed Benchmark Results

This supplement reports detailed benchmark results for all datasets and both labelling strategies (`mvc`, `classic`). Each cell shows mean ± standard deviation over 10 seeds. Columns labelled 0.2, 0.5, and 0.8 refer to target labelling rates $c_{\text{calc}} \in \{0.2, 0.5, 0.8\}$.

1 Benchmark Overview

Table 2 summarizes the best observed F1 score for each dataset and labelling strategy, together with the Naive baseline and the absolute gain. These numbers are the basis for the quantitative statements in the abstract and conclusion.

Table 1 clarifies how the current manuscript extends the earlier non-SCAR study.

Table 1: Comparison of the our earlier non-SCAR study and the present manuscript.

Aspect	**Earlier study**	**Present manuscript**
Task framing	Narrower non-SCAR benchmark centered on clustering and logistic modeling	Broader PU benchmark with six methods and two non-SCAR strategies
Implementation	Initial pipeline and fewer implementation details	Python reimplementation with reproducible preprocessing and evaluation
Experimental scope	Smaller method comparison	Six SpeakLeash corpora, three c_{calc} values, ten seeds
Interpretation	Focus on model behavior	Adds dataset-level discussion, limitations, and practical selection guidance
Reporting	Summary of benchmark behavior	Expanded runtime analysis, stability analysis, and best-F1 summary

Table 2: Best observed F1 score per dataset and strategy, together with the Naive baseline and absolute gain.

Dataset	Strategy	Best F1	Naive F1	Gain
`ISAP_corpus`	classic	0.019	0.000	0.019
`job_offers_pl_corpus`	classic	0.613	0.014	0.599
`open_subtitles_corpus`	classic	0.906	0.861	0.045
`plwiki`	classic	0.771	0.013	0.758
`ulotki_medyczne`	classic	0.833	0.022	0.811
`wolne_lektury_corpus`	classic	0.692	0.365	0.327
`ISAP_corpus`	mvc	0.020	0.000	0.020
`job_offers_pl_corpus`	mvc	0.339	0.004	0.335
`open_subtitles_corpus`	mvc	0.900	0.827	0.073
`plwiki`	mvc	0.716	0.045	0.671
`ulotki_medyczne`	mvc	0.209	0.000	0.209
`wolne_lektury_corpus`	mvc	0.573	0.074	0.499

Table 3: Average wall-clock runtime by method and labelling strategy, aggregated over datasets and target labelling rates.

Method	classic [min]	mvc [min]
Naive	0.003	0.003
Clust	0.026	0.029
Strict-LassClust	4.347	3.809
Non-Strict-LassClust	4.426	4.220
LassoJoint	2.086	2.011
Spy	0.028	0.027

Table 4: Practical method-selection guide derived from the benchmark results.

Data condition	**Preferred method(s)**	**Reason**
Moderately balanced corpus with clear feature signal	`LassoJoint`	Strongest average retrieval quality and good use of sparse informative features
Highly imbalanced corpus or weak propensity signal	`Spy`	More stable when positive support is small and class separation is difficult
Need for a lightweight baseline or quick sanity check	`Clust` / `Naive`	Low runtime and simple behavior make them useful reference points
Need to stress-test label-generation assumptions	Compare `classic` vs `mvc`	Strategy sensitivity is dataset-specific and should be interpreted as part of the benchmark design

2 Labelling Strategy: `mvc`

Table 5: Detailed results for `ISAP_corpus` with labelling strategy `mvc` (mean ± std over 10 seeds).

Method	**AUC**			**PR-AUC**			**F1**		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.089±0.017	0.127±0.018	0.249±0.025	0.005	0.005	0.005	0.000	0.000	0.000
Clust	0.772±0.043	0.104±0.025	0.184±0.030	0.022±0.003	0.005	0.005	0.017	0.000	0.000
S-LassClust	0.820±0.051	0.090±0.021	0.144±0.023	**0.036±0.008**	0.005	0.005	0.017	0.000	0.000
NS-LassClust	**0.821±0.050**	0.090±0.021	0.144±0.023	**0.036±0.008**	0.005	0.005	0.017	0.000	0.000
LassoJoint	0.144±0.036	0.176±0.021	0.241±0.009	0.007	0.005	0.005	0.004±0.001	0.000	0.000
Spy	0.580±0.039	**0.543±0.036**	**0.543±0.036**	0.010±0.001	**0.009±0.001**	**0.009±0.001**	**0.019±0.001**	**0.019**	**0.019**

Table 6: Detailed results for `job_offers_pl_corpus` with labelling strategy `mvc` (mean ± std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	**0.575±0.012**	0.667±0.012	**0.798±0.009**	**0.178±0.005**	0.221±0.008	0.355±0.012	0.000	0.000	0.001±0.002
Clust	0.345±0.012	0.587±0.009	0.663±0.009	0.120±0.002	0.195±0.009	0.256±0.014	**0.287±0.001**	0.040±0.014	0.090±0.021
S-LassClust	0.327±0.006	0.684±0.012	0.758±0.013	0.117±0.001	**0.305±0.020**	**0.383±0.024**	0.286±0.001	0.106±0.021	0.215±0.019
NS-LassClust	0.325±0.008	0.684±0.012	0.758±0.013	0.117±0.001	**0.305±0.020**	**0.383±0.024**	0.286±0.001	0.106±0.021	0.215±0.019
LassoJoint	0.436±0.011	**0.704±0.019**	0.766±0.013	0.158±0.001	0.298±0.020	0.371±0.019	0.116±0.016	0.169±0.023	**0.312±0.017**
Spy	0.484±0.009	0.522±0.011	0.522±0.010	0.154±0.003	0.170±0.005	0.170±0.004	0.032±0.028	**0.274±0.008**	0.271±0.006

Table 7: Detailed results for `open_subtitles_corpus` with labelling strategy `mvc` (mean ± std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	**0.565±0.012**	0.677±0.011	0.819±0.012	**0.682±0.011**	0.761±0.012	0.874±0.012	0.000	0.105±0.009	0.819±0.006
Clust	0.532±0.010	0.488±0.011	0.492±0.010	0.633±0.007	0.604±0.008	0.606±0.007	0.357±0.013	**0.751±0.006**	0.777±0.003
S-LassClust	0.555±0.013	0.521±0.011	0.571±0.010	0.645±0.009	0.627±0.008	0.664±0.009	0.368±0.037	0.746±0.006	0.783±0.004
NS-LassClust	0.558±0.013	0.521±0.011	0.571±0.010	0.648±0.009	0.627±0.008	0.664±0.009	**0.387±0.022**	0.746±0.006	0.783±0.004
LassoJoint	0.510±0.012	**0.826±0.014**	**0.907±0.008**	0.646±0.006	**0.865±0.014**	**0.919±0.009**	0.222±0.043	0.626±0.012	**0.891±0.006**
Spy	0.488±0.005	0.492±0.010	0.490±0.009	0.636±0.002	0.617±0.006	0.617±0.006	0.005±0.003	0.019±0.004	0.018±0.004

Table 8: Detailed results for `plwiki` with labelling strategy `mvc` (mean ± std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.665±0.010	0.643±0.010	0.721±0.008	0.525±0.009	0.517±0.009	0.626±0.013	0.005±0.002	0.023±0.005	0.029±0.008
Clust	0.124±0.005	0.661±0.012	0.706±0.011	0.262±0.001	0.567±0.011	0.626±0.011	0.587±0.001	0.100±0.012	0.122±0.014
S-LassClust	0.153±0.011	0.688±0.011	0.733±0.009	0.266±0.002	0.598±0.011	0.662±0.011	0.588±0.001	0.301±0.013	0.348±0.017
NS-LassClust	0.153±0.011	0.688±0.011	0.733±0.009	0.266±0.002	0.598±0.011	0.662±0.011	0.588±0.001	0.301±0.013	0.348±0.017
LassoJoint	**0.729±0.014**	**0.781±0.009**	**0.853±0.005**	**0.590±0.010**	**0.727±0.012**	**0.812±0.009**	**0.691±0.020**	**0.620±0.015**	**0.643±0.010**
Spy	0.409±0.022	0.482±0.005	0.481±0.005	0.372±0.011	0.408±0.003	0.407±0.002	0.526±0.019	0.579±0.003	0.578±0.003

Table 9: Detailed results for `ulotki_medyczne` with labelling strategy `mvc` (mean ± std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.173±0.009	0.545±0.021	0.827±0.011	0.035	0.065±0.004	0.243±0.020	0.000	0.000	0.000
Clust	0.600±0.012	0.197±0.007	0.282±0.011	0.069±0.002	0.036	0.040±0.001	0.159±0.003	0.001±0.001	0.004±0.002
S-LassClust	**0.677±0.019**	0.262±0.009	0.331±0.011	0.089±0.006	0.039	0.043±0.001	**0.165±0.003**	0.008±0.002	0.013±0.004
NS-LassClust	0.676±0.020	0.262±0.009	0.331±0.011	0.089±0.007	0.039	0.043±0.001	0.164±0.003	0.008±0.002	0.013±0.004
LassoJoint	0.306±0.012	0.799±0.025	**0.951±0.004**	0.062	0.193±0.026	**0.522±0.032**	0.001±0.001	0.000	0.015±0.007
Spy	0.633±0.068	**0.854±0.011**	0.851±0.011	**0.124±0.019**	**0.225±0.012**	0.220±0.013	0.000	**0.159±0.034**	**0.149±0.031**

Table 10: Detailed results for `wolne_lektury_corpus` with labelling strategy `mvc` (mean ± std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	**0.626±0.012**	0.650±0.016	0.718±0.012	**0.416±0.009**	0.431±0.011	**0.517±0.014**	0.000	0.002±0.002	0.054±0.013
Clust	0.261±0.009	0.561±0.014	0.624±0.016	0.261±0.002	0.376±0.008	0.421±0.012	**0.569±0.003**	0.048±0.010	0.081±0.011
S-LassClust	0.298±0.024	0.596±0.012	0.656±0.012	0.270±0.006	0.398±0.008	0.438±0.010	**0.569±0.003**	0.098±0.010	0.168±0.018
NS-LassClust	0.298±0.023	0.596±0.012	0.656±0.012	0.270±0.005	0.398±0.008	0.438±0.010	**0.569±0.003**	0.098±0.010	0.168±0.018
LassoJoint	0.493±0.014	**0.660±0.012**	**0.726±0.011**	0.375±0.006	**0.445±0.010**	0.499±0.013	0.251±0.032	0.157±0.020	0.262±0.017
Spy	0.498±0.012	0.534±0.007	0.533±0.007	0.375±0.006	0.395±0.004	0.394±0.004	0.520±0.012	**0.557±0.004**	**0.556±0.004**

3 Labelling Strategy: `classic`

Table 11: Detailed results for `ISAP_corpus` with labelling strategy `classic` (mean ± std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	**0.826$\pm$0.055**	**0.757$\pm$0.062**	**0.645$\pm$0.058**	**0.038$\pm$0.010**	**0.025$\pm$0.008**	**0.014$\pm$0.003**	0.000	0.000	0.000
Clust	0.640$\pm$0.092	0.598$\pm$0.086	0.491$\pm$0.062	0.012$\pm$0.003	0.011$\pm$0.003	0.008$\pm$0.001	0.000	0.000	0.000
S-LassClust	0.682$\pm$0.068	0.538$\pm$0.052	0.416$\pm$0.046	0.013$\pm$0.002	0.009$\pm$0.001	0.007$\pm$0.001	0.000	0.000	0.000
NS-LassClust	0.682$\pm$0.068	0.558$\pm$0.091	0.416$\pm$0.046	0.013$\pm$0.002	0.010$\pm$0.002	0.007$\pm$0.001	0.000	0.000	0.000
LassoJoint	0.666$\pm$0.077	0.663$\pm$0.081	0.577$\pm$0.042	0.017$\pm$0.008	0.015$\pm$0.005	0.010$\pm$0.001	0.000	0.000	0.000
Spy	0.543$\pm$0.035	0.543$\pm$0.036	0.543$\pm$0.036	0.009$\pm$0.001	0.009$\pm$0.001	0.009$\pm$0.001	**0.019**	**0.019**	**0.019**

Table 12: Detailed results for `job_offers_pl_corpus` with labelling strategy `classic` (mean ± std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.822$\pm$0.011	**0.828$\pm$0.011**	0.819$\pm$0.012	0.377$\pm$0.016	0.384$\pm$0.018	0.373$\pm$0.017	0.000	0.000	0.008$\pm$0.004
Clust	0.662$\pm$0.013	0.713$\pm$0.013	0.730$\pm$0.013	0.267$\pm$0.013	0.309$\pm$0.017	0.322$\pm$0.017	0.039$\pm$0.013	0.068$\pm$0.017	0.117$\pm$0.020
S-LassClust	0.732$\pm$0.015	0.762$\pm$0.037	0.808$\pm$0.013	0.343$\pm$0.016	0.379$\pm$0.052	0.438$\pm$0.023	0.061$\pm$0.015	0.169$\pm$0.099	0.225$\pm$0.017
NS-LassClust	0.732$\pm$0.015	0.781$\pm$0.029	0.808$\pm$0.013	0.343$\pm$0.016	**0.400$\pm$0.040**	0.438$\pm$0.023	0.061$\pm$0.015	0.088$\pm$0.040	0.225$\pm$0.017
LassoJoint	**0.892$\pm$0.008**	0.803$\pm$0.026	**0.888$\pm$0.009**	**0.635$\pm$0.021**	0.398$\pm$0.028	**0.632$\pm$0.020**	**0.590$\pm$0.013**	0.233$\pm$0.201	**0.577$\pm$0.017**
Spy	0.521$\pm$0.010	0.521$\pm$0.010	0.521$\pm$0.010	0.170$\pm$0.004	0.169$\pm$0.004	0.169$\pm$0.004	0.271$\pm$0.006	**0.268$\pm$0.005**	0.267$\pm$0.006

Table 13: Detailed results for `open_subtitles_corpus` with labelling strategy `classic` (mean ± std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	**0.704$\pm$0.023**	**0.852$\pm$0.016**	**0.856$\pm$0.010**	**0.786$\pm$0.019**	**0.893$\pm$0.012**	**0.892$\pm$0.009**	0.004$\pm$0.002	0.131$\pm$0.028	**0.853$\pm$0.005**
Clust	0.482$\pm$0.011	0.486$\pm$0.010	0.490$\pm$0.010	0.600$\pm$0.007	0.603$\pm$0.007	0.606$\pm$0.007	0.741$\pm$0.006	0.772$\pm$0.003	0.780$\pm$0.002
S-LassClust	0.502$\pm$0.012	0.518$\pm$0.026	0.582$\pm$0.009	0.612$\pm$0.009	0.623$\pm$0.018	0.672$\pm$0.010	0.739$\pm$0.005	**0.776$\pm$0.005**	0.785$\pm$0.002
NS-LassClust	0.502$\pm$0.012	0.531$\pm$0.013	0.582$\pm$0.009	0.612$\pm$0.009	0.633$\pm$0.011	0.672$\pm$0.010	0.739$\pm$0.005	0.713$\pm$0.163	0.785$\pm$0.002
LassoJoint	0.469$\pm$0.014	0.846$\pm$0.119	0.508$\pm$0.015	0.604$\pm$0.016	0.879$\pm$0.082	0.641$\pm$0.009	**0.788$\pm$0.002**	0.754$\pm$0.192	0.789$\pm$0.001
Spy	0.490$\pm$0.010	0.489$\pm$0.009	0.488$\pm$0.009	0.617$\pm$0.006	0.617$\pm$0.006	0.617$\pm$0.006	0.018$\pm$0.004	0.018$\pm$0.004	0.018$\pm$0.004

Table 14: Detailed results for `plwiki` with labelling strategy `classic` (mean ± std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.722$\pm$0.023	0.730$\pm$0.012	0.731$\pm$0.009	0.587$\pm$0.035	0.607$\pm$0.020	0.622$\pm$0.016	0.000	0.000$\pm$0.001	0.004$\pm$0.004
Clust	0.710$\pm$0.023	0.705$\pm$0.016	0.720$\pm$0.011	0.654$\pm$0.027	0.638$\pm$0.020	0.648$\pm$0.015	0.011$\pm$0.005	0.019$\pm$0.007	0.045$\pm$0.011
S-LassClust	0.429$\pm$0.034	0.674$\pm$0.050	0.718$\pm$0.011	0.416$\pm$0.027	0.615$\pm$0.055	0.652$\pm$0.015	0.018$\pm$0.007	0.046$\pm$0.023	0.117$\pm$0.015
NS-LassClust	0.429$\pm$0.034	0.643$\pm$0.044	0.718$\pm$0.011	0.416$\pm$0.027	0.577$\pm$0.048	0.652$\pm$0.015	0.018$\pm$0.007	0.030$\pm$0.013	0.117$\pm$0.015
LassoJoint	**0.869$\pm$0.010**	**0.790$\pm$0.065**	**0.881$\pm$0.006**	**0.841$\pm$0.013**	**0.718$\pm$0.105**	**0.851$\pm$0.010**	0.527$\pm$0.054	0.192$\pm$0.165	**0.754$\pm$0.011**
Spy	0.481$\pm$0.005	0.480$\pm$0.005	0.480$\pm$0.005	0.407$\pm$0.002	0.407$\pm$0.002	0.407$\pm$0.002	**0.578$\pm$0.003**	**0.578$\pm$0.003**	0.578$\pm$0.003

Table 15: Detailed results for `ulotki_medyczne` with labelling strategy `classic` (mean ± std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.920±0.004	0.917±0.005	0.915±0.006	0.421±0.022	0.413±0.023	0.406±0.024	0.000	0.000	0.005±0.008
Clust	0.381±0.012	0.427±0.012	0.457±0.010	0.046±0.001	0.050±0.001	0.052±0.001	0.012±0.002	0.016±0.003	0.022±0.005
S-LassClust	0.369±0.013	0.424±0.020	0.462±0.010	0.045±0.001	0.049±0.002	0.053±0.001	0.018±0.004	0.040±0.017	0.042±0.006
NS-LassClust	0.369±0.013	0.438±0.017	0.462±0.010	0.045±0.001	0.051±0.002	0.053±0.001	0.018±0.004	0.022±0.009	0.042±0.006
LassoJoint	**0.989±0.002**	**0.946±0.049**	**0.990±0.002**	**0.782±0.041**	**0.578±0.185**	**0.811±0.040**	**0.782±0.021**	**0.420±0.358**	**0.793±0.020**
Spy	0.849±0.011	0.847±0.011	0.845±0.011	0.217±0.011	0.214±0.011	0.211±0.010	0.148±0.030	0.145±0.032	0.141±0.033

Table 16: Detailed results for `wolne_lektury_corpus` with labelling strategy `classic` (mean ± std over 10 seeds).

Method	AUC			PR-AUC			F1		
	0.2	0.5	0.8	0.2	0.5	0.8	0.2	0.5	0.8
Naive	0.693±0.015	**0.729±0.013**	0.748±0.012	0.478±0.015	**0.523±0.017**	**0.555±0.018**	0.000	0.003±0.002	0.327±0.030
Clust	0.589±0.023	0.643±0.017	0.666±0.016	0.406±0.017	0.437±0.015	0.456±0.015	0.036±0.009	0.058±0.013	0.172±0.026
S-LassClust	0.438±0.033	0.621±0.019	0.664±0.013	0.320±0.014	0.425±0.015	0.458±0.012	0.046±0.011	0.090±0.027	0.279±0.028
NS-LassClust	0.438±0.033	0.621±0.014	0.664±0.013	0.320±0.014	0.424±0.012	0.458±0.012	0.046±0.011	0.078±0.020	0.279±0.028
LassoJoint	**0.767±0.009**	0.709±0.029	**0.776±0.011**	**0.544±0.013**	0.488±0.027	0.554±0.015	**0.601±0.028**	0.299±0.251	**0.681±0.009**
Spy	0.533±0.007	0.534±0.007	0.533±0.007	0.394±0.004	0.394±0.004	0.394±0.004	0.556±0.004	**0.556±0.004**	0.556±0.004

4 Feature Descriptions

Table 17 provides a comprehensive list of all 22 linguistic and stylometric features used in the benchmark. All numeric features are rescaled to the $[0, 1]$ range using min-max normalization before model training.

Table 17: Description of 22 input features (X) used in the benchmark.

Feature	**Description**
`adj_ratio`	Ratio of adjectives to total words
`adjectives`	Count of adjectives in the document
`adverbs`	Count of adverbs in the document
`avg_sentence_length`	Average number of words per sentence
`avg_word_length`	Average number of characters per word
`camel_case`	Count of camelCase words
`capitalized_words`	Count of words starting with uppercase letter
`characters`	Total character count (excluding whitespace)
`gunning_fog`	Gunning Fog index; estimates U.S. school grade needed to understand text
`lexical_density`	Ratio of content words (nouns, verbs, adjectives, adverbs) to total words
`noun_ratio`	Ratio of nouns to total words
`nouns`	Count of nouns in the document
`oovs`	Count of out-of-vocabulary words (not in standard Polish dictionary)
`pos_num`	Count of numerals (number-tagged part-of-speech tokens)
`pos_x`	Count of unknown or rare part-of-speech categories
`punctuations`	Count of punctuation marks
`sentences`	Count of sentences (detected by sentence boundary markers)
`stopwords`	Count of common Polish stopwords (high-frequency, low-content words)
`symbols`	Count of non-alphanumeric symbols (special characters)
`verb_ratio`	Ratio of verbs to total words
`verbs`	Count of verbs in the document
`words`	Total word count

5 Statistical Comparison for F1, AUC, AUCPR for different PU methods with classic and MVC scenarios

Table 18: Results of the paired Wilcoxon signed-rank tests comparing the classic and non-SCAR variants based on the F1 score. For each threshold parameter c, Holm's correction was applied across the six tested methods. Statistically significant adjusted p-values ($\alpha = 0.05$) are shown in bold.

	$c = 0.2$		$c = 0.5$		$c = 0.8$	
Method	W	Holm p	W	Holm p	W	Holm p
Clust	1475	$\mathbf{1.51 \times 10^{-4}}$	445	3.87×10^{-1}	376	5.67×10^{-2}
LassoJoint	210	$\mathbf{1.07 \times 10^{-6}}$	447	4.49×10^{-1}	79.5	$\mathbf{4.42 \times 10^{-7}}$
Naive	120	5.86×10^{-1}	191	1.00	196.5	$\mathbf{3.27 \times 10^{-3}}$
Non-strict LassClust	1475	$\mathbf{1.51 \times 10^{-4}}$	878.5	1.04×10^{-1}	486.5	4.24×10^{-1}
Spy	21	$\mathbf{1.43 \times 10^{-9}}$	801	$\mathbf{8.39 \times 10^{-7}}$	422	$\mathbf{4.49 \times 10^{-5}}$
Strict LassClust	1475	$\mathbf{1.51 \times 10^{-4}}$	586.5	1.00	486.5	4.24×10^{-1}

Table 19: Results of the paired Wilcoxon signed-rank tests comparing the classic and non-SCAR variants based on the AUCPR score. For each threshold parameter c, Holm's correction was applied across the six tested methods. Statistically significant adjusted p-values ($\alpha = 0.05$) are shown in bold.

	$c = 0.2$		$c = 0.5$		$c = 0.8$	
Method	W	Holm p	W	Holm p	W	Holm p
Clust	465	$\mathbf{2.81 \times 10^{-3}}$	45	$\mathbf{1.16 \times 10^{-9}}$	7	$\mathbf{6.66 \times 10^{-10}}$
LassoJoint	152	$\mathbf{9.93 \times 10^{-8}}$	313	$\mathbf{2.85 \times 10^{-5}}$	460	$\mathbf{8.20 \times 10^{-4}}$
Naive	0	$\mathbf{1.00 \times 10^{-10}}$	0	$\mathbf{1.00 \times 10^{-10}}$	114	$\mathbf{3.01 \times 10^{-8}}$
Non-strict LassClust	540	$\mathbf{1.07 \times 10^{-2}}$	338.5	$\mathbf{4.32 \times 10^{-5}}$	246	$\mathbf{3.42 \times 10^{-6}}$
Spy	277	$\mathbf{5.14 \times 10^{-5}}$	402	$\mathbf{5.77 \times 10^{-5}}$	235	$\mathbf{7.45 \times 10^{-4}}$
Strict LassClust	536	$\mathbf{1.07 \times 10^{-2}}$	279	$\mathbf{1.16 \times 10^{-5}}$	246	$\mathbf{3.42 \times 10^{-6}}$

Table 20: Results of the paired Wilcoxon signed-rank tests comparing the classic and non-SCAR variants based on the AUC score. For each threshold parameter c, Holm's correction was applied across the six tested methods. Statistically significant adjusted p-values ($\alpha = 0.05$) are shown in bold.

	$c = 0.2$		$c = 0.5$		$c = 0.8$	
Method	W	Holm p	W	Holm p	W	Holm p
Clust	464	$\mathbf{2.74 \times 10^{-3}}$	45	$\mathbf{1.17 \times 10^{-9}}$	33.5	$\mathbf{1.00 \times 10^{-9}}$
LassoJoint	55	$\mathbf{1.25 \times 10^{-9}}$	189	$\mathbf{2.77 \times 10^{-7}}$	546	$\mathbf{6.67 \times 10^{-3}}$
Naive	0	$\mathbf{1.00 \times 10^{-10}}$	0	$\mathbf{1.00 \times 10^{-10}}$	4	$\mathbf{1.23 \times 10^{-10}}$
Non-strict LassClust	699	2.12×10^{-1}	222.5	$\mathbf{8.06 \times 10^{-7}}$	188	$\mathbf{5.61 \times 10^{-7}}$
Spy	289	$\mathbf{1.65 \times 10^{-5}}$	1036	$\mathbf{2.05 \times 10^{-7}}$	1012	$\mathbf{5.61 \times 10^{-7}}$
Strict LassClust	695	2.12×10^{-1}	200.5	$\mathbf{8.06 \times 10^{-7}}$	187	$\mathbf{5.61 \times 10^{-7}}$